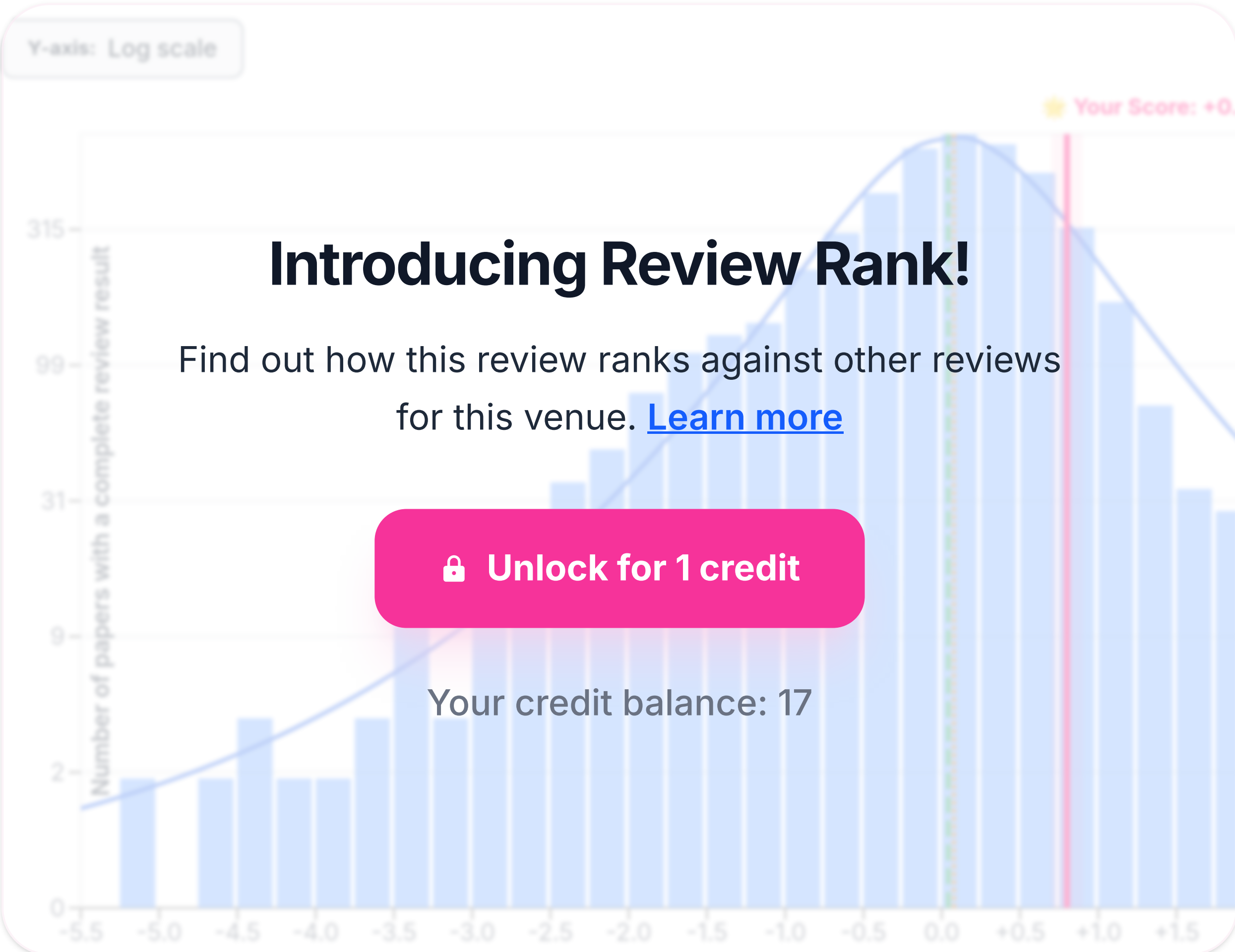
The U-Shaped Challenge of Language Grounding: How Instruction Granularity Induces VLA Shortcuts

Anonymized Authors

View Paper Copy

Reviewed as if submitted to CONFERENCE ACL 2026 - main track



Introducing Review Rank!

Find out how this review ranks against other reviews for this venue. Learn more

Unlock for 1 credit

Your credit balance: 17

Desk Rejection Assessment:

Paper Length

Pass ✅.

Topic Compatibility

Pass ✅. The paper studies language grounding and instruction following in multimodal Vision-Language-Action (VLA) agents, introduces a benchmark, and analyzes model behavior. This is squarely within "Multimodality and Language Grounding to Vision, Robotics and Beyond" and "Resources and Evaluation" for ACL.

Minimum Quality

Pass ✅. The submission is complete and self-contained: Abstract, Introduction, Related Work, Method (Sections 3.1–3.4), Benchmark (Section 4), Experiments (Section 5) with quantitative analyses and multiple figures, Conclusion, and a Limitations section. Claims are mostly supported by experiments; while there are issues to discuss, nothing rises to desk-reject severity.

Prompt Injection and Hidden Manipulation Detection

Pass ✅. No indications of hidden prompts or manipulative content; the text uses standard scholarly structure.
(if any assessment indicate a possible desk rejection, STOP here and DO NOT include any sections below)

Expected Review Outcome:

Paper Summary

The paper proposes an agent-centric metric of instruction granularity, rule width (w), adapted from width-based planning, defined as the maximum number of state features that must be tracked concurrently to fulfill a rule in an instruction set. It introduces Mini-BEHAVIOR-Gran, a controlled benchmark over 20 tasks where instructions are generated from rule sets with varying widths while perception and control are held constant. Using three VLA action-decoding heads on a shared VLM backbone, the authors report a U-shaped success-rate trend across instruction granularity: high performance at w=0 (fine) and an unexpected rebound at coarse (w≥2), followed by collapse at w≥4. They attribute the rebound to "language-ignoring" shortcuts driven by higher predictability P(l|v) for coarse texts, demonstrate asymmetric generalization (coarse→fine transfers better than fine→coarse), and show mixed-granularity (Axial) training reduces modality collapse.

Summary Of Strengths

- Conceptual framing: Treating instruction granularity as an agent-centric planning burden rather than surface linguistic complexity is well motivated. The adoption of rule width as a cross-task, capacity-like measure is interesting and plausibly connects language to planning complexity (Section 3.4).
- Benchmark: Mini-BEHAVIOR-Gran is a useful controlled setup that manipulates instruction granularity while fixing perception/control, supported by an Oracle Instructor to avoid instruction applicability confounds (Figure 2).
- Clear empirical narrative:
  - Figure 4 (width vs instruction horizon) shows SR monotonically degrades with width even at fixed rule-call bins, supporting w as a difficulty proxy.
  - Figure 5 shows the non-monotonic U-shaped trend in matched train/test granularity across three distinct action decoders, suggesting the finding is architectural-agnostic.
  - Figure 6 quantifies that coarse texts are more predictable from vision (lower perplexity), consistent with the hypothesized $P(l|v)$ effect.
  - Figure 7 (ΔSR with vs without language) backs the modality-collapse diagnosis and the mitigation via mixed granularity training.
  - Figure 9 shows the training–test width transfer asymmetry and an apparent collapse at $w \geq 4$.
- Methodological clarity and examples: Table 1 and Figure 1 concretely illustrate how width accrues in the “cleaning k shoes” task and how instruction factoring reduces effective width, giving readers an interpretable anchor for the definition.
- Breadth of evaluation: Comparison across three representative action-decoding strategies (autoregressive, discrete diffusion, and parallel decoding) strengthens generality of the observations; analysis of failure modes (regression vs stagnation) is helpful (Figure 9 right panel).

Summary Of Weaknesses

1. Definition and computation of width for natural language rules are partly informal and may be subjective.
   - Section 3.4 defines rule width as the minimal conjunction size required by novelty-driven BFS, and then argues that previously satisfied features (e.g., Hr from r1a) are treated as “maintained,” effectively lowering width for r1b. However, the conversion from natural-language templates to rule schemas and then to an operational w sometimes depends on designer choices about which features are “tracked” vs “maintained.” A concrete, algorithmic procedure to derive w from a natural-language instruction is not fully specified, and the “maintained feature” allowance risks post-hoc assignment to fit intuition. The paper would benefit from an explicit algorithm, including edge cases such as loops, negation, or non-Markovian clauses.

2. Potential circularity and external validity of the Oracle Instructor setup.
   - While the Oracle Instructor (Figure 2) ensures rule applicability, dynamic instruction emission conditioned on current state is not representative of many real-world VLA deployments where instructions are static or only sparsely updated. This could make language easier to align than in-the-wild settings, and it couples success to the particular rule segmentation. The authors partially acknowledge this, but the implications on generalizability and whether the U-shape persists with fixed instructions are not explored. A “static-instruction” ablation in the main paper would make the claims stronger.
3. Evidence for the predictability hypothesis is suggestive but not conclusive.
   - Figure 6 shows small absolute differences in perplexity across granularities (≈1.081→1.048). There are no error bars, confidence intervals, or alternative proxies (e.g., mutual information estimates or controlled class-imbalance checks). Without significance testing, it is hard to conclude these small gaps are the main driver. Similarly, Figure 8’s attention analysis shows no reduction in language-token attention, which weakens the mechanistic link between $P(l|v)$, attention, and policy invariance. Additional probes (e.g., causal interventions on text tokens, counterfactual masking beyond a “static goal” baseline) would strengthen the claim.
4. Scope and reproducibility gaps in the main text.
   - Several essential details are deferred to appendices (e.g., §E, §G), including planner settings, data splits, hyperparameters, seeds, and whether code/rules will be released. For a benchmark-centric paper, clear in-paper specifications of the dataset construction protocol and release plan are important to ensure adoption and verify w assignments per task. At minimum, the main paper should include a concise, explicit description of how Φ and rule sets were designed for all 20 tasks and how w was validated (soundness/closure checks are only asserted on Page 4).
5. Threats to validity in the “capacity cap at w≥4.”
   - Figure 9 (left heatmap) reports near-uniform failure at w≥4 across training setups, but the text does not disclose how many tasks populate that bin nor whether action-space or environment idiosyncrasies contribute. Since Φ and rules are handcrafted, high-width bins may be under-sampled or contain especially tricky spatial preconditions. A per-task table summarizing width distribution, counts, and success would help validate that the observed cap stems from width, not dataset skew. Right now the cap might conflate width and rare-task effects.

Additional figure/table-specific comments:

- Figure 4 is one of the strongest pieces of evidence: by binning rule calls, it visually separates horizon from width. However, the “nan%” cells complicate interpretation; clarifying sample sizes per cell or gray-out for “no data” would avoid misreading zeros as failures.

- Table 1 nicely demonstrates width growth across “Cleaning k shoes,” but it silently treats preservation of C1 while cleaning C2 as a hard concurrency, which assumes no reversible dynamics. If the environment allowed re-soiling or accidental drops, width could shift. An explicit statement of the invariants used in width computation would help align Table 1 with the formal definition.
- Figure 5 (U-shape) is a central result. Please include standard deviations over seeds. Without dispersion, it is difficult to judge whether differences among decoders or granularities are statistically meaningful.

Potentially Missing Related Work

1. Nguyen, V.-Q., Suganuma, M., Okatani, T., 2021 — Look Wide and Interpret Twice: Improving Performance on Interactive Instruction-following Tasks. Relevance: addresses staged interpretation of instructions, related to granularity and decomposition. Suggest discussing in Section 2 as complementary strategies to reduce planning burden, and relate to rule width after Eq. (def. of w in Section 3.4).
2. Huang, W., Xia, F., Xiao, T., 2022 — Inner Monologue: Embodied Reasoning through Planning with Language Models. Relevance: language-based planning for embodied agents; discuss how rule width compares with their planning abstractions. Add to Related Work and contrast in Section 5’s discussion.
3. Goel, D., Singh, K. P., Choi, J., 2022 — Language Guided Meta-Control for Embodied Instruction Following. Relevance: meta-control with language to structure tasks; connect in Section 2 and consider as a baseline frame in discussion.
4. Grazioso, M., Suglia, A., 2023 — An Analysis of Visually Grounded Instructions in Embodied AI Tasks. Relevance: analyzes instruction properties and their grounding; useful context for your critique of text-centric measures. Cite in Section 2 to better situate the paper.
5. Körner, A., Castillo, M., Drijvers, L., 2023 — Embodied Processing at Six Linguistic Granularity Levels: A Consensus Paper. Relevance: taxonomy of linguistic granularity; compare to agent-centric width and clarify distinctions in Section 1–2.
6. Li, D., Zhao, C., Yang, S., 2024 — Learning Instruction-Guided Manipulation Affordance via Large Models. Relevance: instruction-conditioned affordance, connecting feature-level constraints to actions. Cite in Related Work.
7. Wang, R., Yu, T., Wu, J., 2024 — Weakly-supervised VLM-guided Partial Contrastive Learning for Visual Language Navigation. Relevance: addresses vision–language alignment in navigation; the analysis of language predictability could be connected in Section 5.4.
8. Zhang, S., Zhang, J., Liu, J., 2024 — Training Language Model Agents without Modifying Language Models. Relevance: agent training paradigms that could

interact with granularity; discuss in Section 5.5 as an alternative training lens.

9. Pugacheva, D., Moskalenko, A., Shepelev, D., 2025 — Bring the Apple, Not the Sofa: Impact of Irrelevant Context in Embodied AI Commands on VLA Models. Relevance: robustness to irrelevant or coarse context; compare outcomes with your language-ignoring phenomenon in Section 5.4.
10. Gao, J., Ye, F., Zhang, J., 2025 — Compressor-VLA: Instruction-Guided Visual Token Compression for Efficient Robotic Manipulation. Relevance: instruction-guided visual abstraction; discuss interplay with rule width in Section 6.

Comments Suggestions And Typos

Actionable questions/suggestions for rebuttal:

1. Formalization of width from language: Provide an explicit algorithm to compute w($\mathscr{R}$) from natural-language rules, including how “maintained” features are identified and how loops/negations are handled. If this is currently human-annotated, quantify inter-annotator agreement on w across several tasks.
2. U-shape robustness: Report seed variances and significance tests for Figures 4–7 and 9. Add per-cell sample counts to heatmaps. Include an ablation with fixed, static instructions (no Oracle updating) to test whether the U-shape and the cap at w≥4 persist.
3. Predictability analysis: Complement Figure 6 with alternative measures (e.g., normalized mutual information between visuals and instruction labels across granularities; control for class imbalance). Add causal masking experiments where specific instruction tokens are ablated or contradicted to quantify reliance beyond ΔSR.
4. Capacity cap: Provide a per-task table summarizing the width distribution and number of episodes at w≥4, plus success by task. If possible, add an explicit memory-augmented baseline to test whether the cap is architectural (lack of memory) rather than inherent to width.
5. Reproducibility: In the camera-ready, include a compact specification of Φ for each of the 20 tasks and publish the rule sets and the generator templates. Clarify licensing and release timeline. State hardware, seeds, hyperparameters, and BFWS settings in the main paper.

Typos and minor issues:

- Page 3, Table 1: multiple parentheses mismatches in the “Concurrent Features Tracked” column (e.g., “(px = X), ..., (pr = 0)]”; “pft ↓)”); please standardize.
- Page 4: “Width as a Planning Proxy.:” double punctuation.
- Page 6: “The primary architectural difference among latest VLA models...” → “among the latest...”.
- Page 7: “horis setup. Transfer is asymmetric: train coarse → test fine works, but zon (e.g., laying woods) tends to cause re gression” — figure caption seems truncated/garbled; please fix wording and spacing.

- Several places: inconsistencies in variable typesetting (H_r vs Hr), and sink/tap naming. Please standardize.

Criteria that could raise my score:

- A clear algorithm for computing w from language with agreement metrics.
- Stronger statistical support and ablations for the P(I|v) hypothesis and the U-shaped trend under static instructions.
- Per-task reporting for high-width bins and an additional memory-augmented baseline probing the cap.

Confidence

4 Quite sure after careful checking: I tried to check the important points carefully. It's unlikely, though conceivable, that I missed something that should affect my ratings.

Soundness

3.5 Between Acceptable and Strong: The study is generally sound and thorough, but several core claims (predictability driver; hard cap at w≥4) need stronger statistical support and reporting.

Excitement

3.0 Interesting: I might mention some points of this paper to others and/or attend its presentation in a conference if there's time.

Overall Assessment 📊 See how you rank ↑

3.5 Borderline Conference: The paper is close to being acceptable for the main conference but has notable weaknesses. Justification: The agent-centric formalization of granularity via width and the controlled benchmark are useful contributions, and the reported U-shaped phenomenon with cross-architecture replication is interesting. However, the definition and computation of width from language are partly subjective, the predictability mechanism is only weakly evidenced, and the capacity-cap claim needs per-task accounting and stronger statistics. With the suggested clarifications and ablations, this could solidify into a strong conference paper; as is, it sits on the border.

Best Paper Justification

N/A.

Limitations And Societal Impact

The paper includes a dedicated Limitations section (Section 7) acknowledging discrete action spaces, predefined feature sets Φ, and static visual granularity. Additional societal impact concerns are minimal because experiments are simulation-only and do not involve human data. Still, clarifying how the benchmark avoids encoding biased object–action priors in templates would be useful, as language predictability could entrench dataset-specific shortcuts.

Ethical concerns

None.

Needs Ethics Review

No.

Reproducibility

4 Mostly reproducible: Main components are described and figures provide trends; however, release of Φ, rule sets, and templates is essential to ensure exact replication.

Datasets

3 Potentially useful: If released with full rule sets and templates, Mini-BEHAVIOR-Gran could be useful for controlled studies of instruction granularity.

Software

3 Potentially useful.: The Oracle Instructor, rule generators, and evaluation harness would be useful if released; details in the main text are not yet sufficient on their own.
======================== Below are simulated alternative reviewer personas covering the full range of Overall Assessment scores. These are provided to aid ACs/SACs in triangulating decisions; they are not additional votes. — Reviewer persona (Overall = 1.0 Do not resubmit) — Paper Summary: Same as above. Strengths: Interesting angle; clear illustrations (Table 1, Figure 1).
Weaknesses:

- Fundamental subjectivity: No algorithmic, reproducible procedure to map language to width; designer-dependent choices undermine the central claim (Section 3.4).
- Oracle Instructor makes the setting unrealistic and possibly circular; dynamic rules steer the agent, conflating instruction adherence with state-conditioned prompting (Figure 2).

- Predictability claim rests on tiny PPL differences (Figure 6) without statistics; ΔSR (Figure 7) can reflect implementation details of "without language" rather than ignoring.
- No statistically robust evidence of U-shape (Figure 5) or cap (Figure 9): no error bars, seeds, or per-task counts.
- Reproducibility weak: Φ and rules for 20 tasks not disclosed in main paper. Figure/Table: Figure 4's "nan%" cells and lack of counts render its interpretation suspect; Table 1 assumes invariants that are unstated. Confidence: 4 Soundness: 1.5 Between Major Issues and Poor Excitement: 1.5 Slightly below potentially interesting. Overall: 1.0 Do not resubmit Reproducibility: 2 Datasets: 2 Software: 2 Ethics: None

— Reviewer persona (Overall = 1.5 Resubmit after next cycle) — Summary: Same. Strengths: Benchmark direction is promising; Figure 5's cross-decoder U-shape is intriguing. Weaknesses: As above, but suggests fixable paths:

- Formal width-computation algorithm and IAA.
- Static-instruction ablation and statistics for Figures 4–7,9.
- Per-task counts for w≥4. Figure/Table: Table 1 is clear but needs invariants stated; Figure 2's reliance on Oracle undermines external validity. Confidence: 4 Soundness: 2 Excitement: 2 Overall: 1.5 Resubmit after next cycle Reproducibility: 3 Datasets: 3 Software: 3 Ethics: None

— Reviewer persona (Overall = 2.0 Resubmit next cycle) — Summary: Same. Strengths: Agent-centric metric; nice visuals; breadth across three decoders. Weaknesses: Core claims under-evidenced; subjectivity in w; limited external validity; missing statistics; insufficient release details. Figure/Table: Figure 5 needs error bars; Figure 4 needs counts; Table 1 good but assumptions unclear. Confidence: 4 Soundness: 2.5 Between Poor and Acceptable Excitement: 2 Overall: 2.0 Resubmit next cycle Reproducibility: 3 Datasets: 3 Software: 3 Ethics: None

— Reviewer persona (Overall = 2.5 Borderline Findings) — Summary: Same. Strengths: Clean framing; useful benchmark idea; several consistent patterns across models (Figures 4–5, 7). Weaknesses: Needs tighter formalization and statistical treatment; capacity-cap claim unconvincing without per-task breakdown; Oracle setting may inflate results. Figure/Table: Strongest evidence is Figure 4; still needs sample sizes. Table 1 helpful but clarify maintained-features logic. Confidence: 4 Soundness: 3 Excitement: 2.5 Overall: 2.5 Borderline Findings Reproducibility: 3 Datasets: 3 Software: 3 Ethics: None

— Reviewer persona (Overall = 3.0 Findings) — Summary: Same. Strengths: Width-as-granularity is a clear, inspectable knob; solid qualitative diagnosis with ΔSR (Figure 7) and failure modes (Figure 9 right). Weaknesses: Statistics/ablation gaps; subjectivity in computing w; generalization claims hinge on Oracle; limited per-task transparency in high-width bins. Figure/Table: Figure 5 shows U-shape consistently; add dispersion. Table 1 exemplifies width scaling well. Confidence: 4

Soundness: 3 Excitement: 3 Overall: 3.0 Findings Reproducibility: 4 Datasets: 3 Software: 3 Ethics: None

— Reviewer persona (Overall = 3.5 Borderline Conference) — [This is the primary review above.]

— Reviewer persona (Overall = 4.0 Conference) — Summary: Same. Strengths:

- A principled, cross-task operationalization of instruction granularity via rule width that moves past text-centric proxies;
- A carefully controlled benchmark with Oracle Instructor to isolate granularity;
- Strongly replicated qualitative phenomena across diverse action heads (Figures 4–7, 9);
- Concrete examples (Table 1, Figure 1) that clarify the formalism;
- Practical takeaway: mixed-granularity training improves robustness. Weaknesses: Lacks statistical rigor and an explicit width-derivation procedure; external validity under static instructions not proven; high-width cap needs per-task counts. Figure/Table: Figure 4 powerfully decouples horizon from width; Table 1 is a helpful tutorial device. Confidence: 4 Soundness: 4 Excitement: 3.5 Overall: 4.0 Conference Reproducibility: 4 Datasets: 4 Software: 4 Ethics: None

— Reviewer persona (Overall = 4.5 Borderline Award) — Summary: Same. Strengths: Elegant formal lens (width) that unifies disparate observations about VLA language sensitivity vs ignoring; benchmark and cross-architecture replication; diagnostic analyses (ΔSR, failure modes) with actionable prescription (Axial training). Weaknesses: Statistical rigor and per-task transparency still lacking; width computation needs algorithmic clarity. Figure/Table: Figure 5 is compelling across three decoders; Figure 7 gives a crisp operationalization of language ignoring; Table 1 is pedagogical. Confidence: 4 Soundness: 4.5 Excitement: 4.5 Overall: 4.5 Borderline Award Reproducibility: 4 Datasets: 4 Software: 4 Ethics: None

— Reviewer persona (Overall = 5.0 Consider for Award) — Summary: Same. Strengths: Foundational framing of instruction granularity with strong empirical validation and a reusable benchmark; insights likely to shape how the community designs and trains VLAs. Weaknesses: Minor reporting gaps only. Figure/Table: Figures 4–7,9 and Table 1 triangulate the story comprehensively. Confidence: 4 Soundness: 5 Excitement: 5 Overall: 5.0 Consider for Award Reproducibility: 4 Datasets: 5 Software: 5 Ethics: None

====================== Selection and synthesis rationale:

- Review0 (adopted): Overall = 3.5 (Borderline Conference). Best-aligned and balanced given contributions and current evidence.
- Review-1: Overall = 3.0 (Findings). One level lower, emphasizing the same issues with a more conservative stance.
- Review+1: Overall = 4.0 (Conference). One level higher, reflecting a more optimistic read while acknowledging remaining gaps.

To align with the score calibration rules and avoid over-concentration at a borderline rating, I maintain the 3.5 recommendation but note that stronger statistics and a concrete width-computation algorithm could cleanly lift this to 4.0.

Reviewed on March 09, 2026 GPT-5